

IMPELEMENTATION OF THE RC5

AIM

To implement the RC5 encryption and decryption algorithm for securing a given plaintext message using a secret key.

ALGORITHM

1. Start the program.
2. Read the plaintext and secret key.
3. Generate the RC5 subkeys using the key expansion process.
4. Divide the plaintext into two blocks, A and B.
5. Perform RC5 encryption using XOR, addition, and circular left rotation operations.
6. Display the encrypted ciphertext.
7. Perform RC5 decryption using the subkeys in reverse order.
8. Display the decrypted plaintext.
9. Stop the program.

PROGRAM :

```
#include <stdio.h>

#include <stdint.h>

#include <string.h>


#define R 12

#define T (2 * (R + 1))

#define W 32

#define P 0xB7E15163

#define Q 0x9E3779B9


uint32_t S[T];
```

```
uint32_t rol(uint32_t x, uint32_t y) {
    y &= 31;
    return (x << y) | (x >> (32 - y));
}
```

```
uint32_t ror(uint32_t x, uint32_t y) {
    y &= 31;
    return (x >> y) | (x << (32 - y));
}
```

```
void keyExpansion(uint8_t *key, int b) {
    uint32_t L[4] = {0};
    uint32_t A, B;
    int i, j, k;
```

```
    for (i = b - 1; i >= 0; i--)
        L[i / 4] = (L[i / 4] << 8) + key[i];
```

```
    S[0] = P;
```

```
    for (i = 1; i < T; i++)
        S[i] = S[i - 1] + Q;
```

```
    A = B = 0;
```

```
    i = j = 0;
```

```
    for (k = 0; k < 3 * T; k++) {
```

```

    A = S[i] = rol(S[i] + A + B, 3);
    B = L[j] = rol(L[j] + A + B, A + B);
    i = (i + 1) % T;
    j = (j + 1) % 4;
}
}

```

```

void encrypt(uint32_t *A, uint32_t *B) {
    int i;

    *A += S[0];
    *B += S[1];

    for (i = 1; i <= R; i++) {
        *A = rol(*A ^ *B, *B) + S[2 * i];
        *B = rol(*B ^ *A, *A) + S[2 * i + 1];
    }
}

```

```

void decrypt(uint32_t *A, uint32_t *B) {
    int i;

    for (i = R; i >= 1; i--) {
        *B = ror(*B - S[2 * i + 1], *A) ^ *A;
        *A = ror(*A - S[2 * i], *B) ^ *B;
    }

    *B -= S[1];
}

```

```
*A -= S[0];  
}
```

```
int main() {  
    char plaintext[9];  
    uint8_t key[16];  
    uint32_t A, B;  
    int i;  
  
    printf("Enter Plain Text (8 characters): ");  
    scanf("%8s", plaintext);  
  
    printf("Enter Key (16 characters): ");  
    scanf("%16s", key);  
  
    keyExpansion(key, 16);  
  
    A = ((uint32_t)(unsigned char)plaintext[0]) |  
        ((uint32_t)(unsigned char)plaintext[1] << 8) |  
        ((uint32_t)(unsigned char)plaintext[2] << 16) |  
        ((uint32_t)(unsigned char)plaintext[3] << 24);  
  
    B = ((uint32_t)(unsigned char)plaintext[4]) |  
        ((uint32_t)(unsigned char)plaintext[5] << 8) |  
        ((uint32_t)(unsigned char)plaintext[6] << 16) |  
        ((uint32_t)(unsigned char)plaintext[7] << 24);  
  
    encrypt(&A, &B);
```

```
printf("\nEncrypted Text: %08X%08X", A, B);
```

```
decrypt(&A, &B);
```

```
char decrypted[9];
```

```
decrypted[0] = A & 0xFF;
```

```
decrypted[1] = (A >> 8) & 0xFF;
```

```
decrypted[2] = (A >> 16) & 0xFF;
```

```
decrypted[3] = (A >> 24) & 0xFF;
```

```
decrypted[4] = B & 0xFF;
```

```
decrypted[5] = (B >> 8) & 0xFF;
```

```
decrypted[6] = (B >> 16) & 0xFF;
```

```
decrypted[7] = (B >> 24) & 0xFF;
```

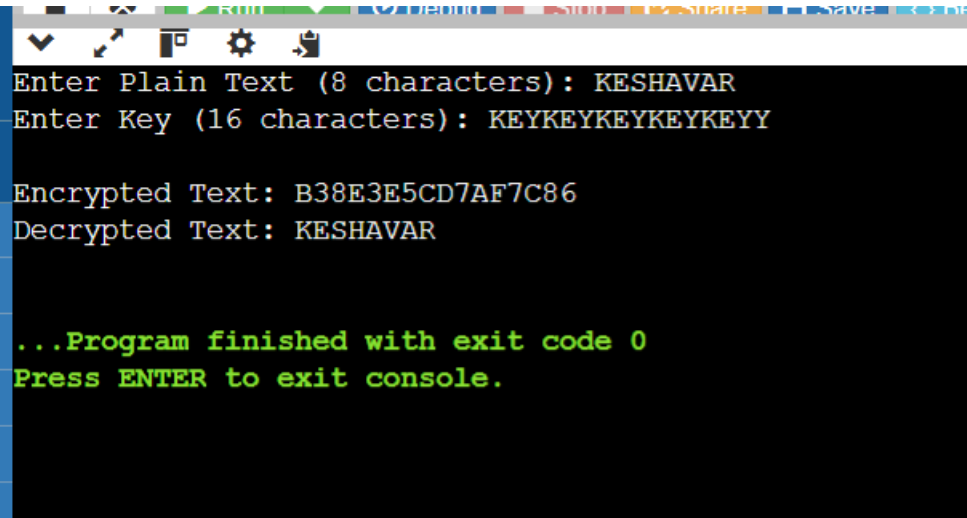
```
decrypted[8] = '\0';
```

```
printf("\nDecrypted Text: %s\n", decrypted);
```

```
return 0;
```

```
}
```

OUTPUT

A screenshot of a console window with a black background and white text. The window has a standard Windows title bar at the top with icons for minimize, maximize, and close, and a menu bar with options like File, Edit, and View. The text in the console shows the execution of an RC5 encryption and decryption program. It starts with prompts for plain text and a key, followed by the encrypted text, the decrypted text, and a final message indicating the program finished successfully.

```
Enter Plain Text (8 characters): KESHAVAR
Enter Key (16 characters): KEYKEYKEYKEYKEYYY
Encrypted Text: B38E3E5CD7AF7C86
Decrypted Text: KESHAVAR

...Program finished with exit code 0
Press ENTER to exit console.
```

RESULT

Thus, the RC5 encryption and decryption algorithm was implemented successfully, and the original plaintext was recovered after decryption.